

Resumes that pass two readers

Week 2, Lecture 1 · Landing the Offer: SWE Job Search for Senior CS Students

Autumn 2026

What we will cover

- How ATS keyword screens work and why they filter before any human sees your resume
- What a recruiter does in the first 30 seconds with a resume
- How to write accomplishment bullets that signal impact, scope, and ownership
- Length, structure, and common formatting traps

Part 1: the ATS screen

Two readers, two very different goals

- Reader 1: an ATS (Applicant Tracking System) that scores keyword density
- Reader 2: a recruiter who decides in roughly 30 seconds whether to pass you to a hiring manager
- Both must say yes before anyone technical sees your work
- Most resumes fail the first reader without the author realizing it

What an ATS actually does

- Parses your resume into structured fields: skills, titles, dates, education
- Scores keyword overlap against the job description
- Resumes below the threshold are often never opened by a human
- Common failure modes: non-standard section headers, tables, columns, graphics, PDFs with unselectable text

Keep the format simple. ATS parsers are not as smart as they look. (Tay, Tech Interview Handbook, 2023)

Matching keywords without gaming the system

- Read the job description carefully; copy the exact technology names it uses
- If the JD says "React" and you write "ReactJS," a naive parser may treat them differently
- List technologies in a dedicated "Skills" section AND weave them into bullet context
- Do not stuff keywords in white text or hidden sections: modern ATS flag this and it destroys trust if a human notices

Auditing your keyword gaps

Three-step audit:

1. Paste the job description into a plain text editor; highlight every technology and verb
2. Compare the list against your resume line by line
3. Any keyword that appears in the JD but not your resume is a gap to address honestly

The top three gaps are the highest-leverage edits you can make before submitting. (Tay, 2023)

Part 2: the 30-second human read

What a recruiter sees in 30 seconds

- Name and contact info: present or not
- Education: school, degree, graduation date
- Most recent role or project title
- Two or three bullet points near the top
- The visual structure: is this easy to scan?

If any of these fails, the resume goes to the "maybe later" pile, which is usually a rejection pile.

Visual hierarchy earns the read

- Use consistent date alignment (flush right, every entry)
- Bold company or project names, not random words in bullets
- One font, two weights (regular and bold), one size range (10-12pt body, 14pt name)
- White space is not wasted space: dense pages signal poor editing, not more experience

Tay (2023): recruiters at high-volume companies spend closer to 10-15 seconds on a first pass. Every second costs.

The top third of the page is prime real estate

- Your most recent and most relevant experience goes first
- A summary section is optional for new grads; skip it if you cannot fill it with specifics
- Education goes at the top only while you are a student or recent grad with no full-time experience
- Projects are first-class citizens when you have no internship or full-time experience to anchor them

Part 3: accomplishment bullets

Task bullets vs. accomplishment bullets

Task bullet (weak)

Accomplishment bullet (strong)

- Worked on the backend API for a web application
 - Helped with database optimization
 - Wrote unit tests for the payment module
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- Built a REST API in Go that reduced average response time from 420ms to 85ms, supporting 3x the original traffic
 - Cut slow-query count by 60% by adding composite indexes; identified gaps with EXPLAIN output

The formula behind a strong bullet

Every bullet answers three questions:

1. **What did you do?** (action verb, specific technology)
2. **At what scale?** (team size, data volume, users affected, code lines, time pressure)
3. **With what result?** (metric before and after, or a concrete outcome)

If you cannot answer all three, the bullet is describing a task, not an accomplishment.

Choosing your action verb

- Lead with a verb that names the work: Built, Designed, Implemented, Optimized, Reduced, Shipped, Refactored
- Avoid: Helped, Assisted, Worked on, Was responsible for, Contributed to
- The verb signals ownership. "Implemented" and "Helped implement" are not the same seniority signal
- Inoue (2020): "wrote a unit test suite that increased code coverage from 2% to 40% and reduced the bug backlog by 50%." The verb is the first word; the metric is the punch.

When you do not have a metric

Not every project has a production dashboard. Still quantify where you can:

- Team size and role ("sole backend engineer on a 4-person team")
- Time pressure ("shipped in 3 weeks for a course demo deadline")
- Scale of the thing built ("processed 50k rows per batch run")
- Adoption ("deployed to 200 students in the pilot cohort")

A scoped claim beats a vague one. A vague claim beats silence.

Part 4: length and structure

One page for new grads, always

- One page is the industry standard for candidates with under 3 years of experience
- A two-page resume from a new grad signals poor editing, not depth
- If you cannot fit your work in one page, cut weaker bullets, not the page limit
- Exception: some academic or research-track roles expect a CV; know which you are applying to

Section order that works

Recommended order for a new-grad SWE resume:

1. Name and contact (GitHub, LinkedIn, email, city/state only)
2. Education (degree, school, graduation date, relevant coursework optional)
3. Experience (internships, research, part-time roles, in reverse chronological order)
4. Projects (2-4, with the strongest first)
5. Skills (languages, frameworks, tools; no proficiency bars)

Only include achievements related to the job application and try to quantify your achievements. Highlighting a few of your best achievements is better than including many 'average' achievements in your resume.

<footer>Yangshun Tay, Tech Interview Handbook, 2023</footer>

Take-aways

1. An ATS reads before any human does. Use the job description's exact technology names, keep formatting simple, and audit for keyword gaps.
2. The 30-second human read starts at the top third of the page. Visual hierarchy earns the right to be read.
3. Every bullet should answer: what did you do, at what scale, with what result. Task descriptions are not accomplishments.
4. Lead with an ownership verb. "Implemented" and "helped implement" send very different seniority signals.
5. One page for new grads, section order matters, and skills bars add no information.

What's next

- **Reading:** Week 2 reading covers ATS auditing, bullet rewrites, and GitHub/LinkedIn in one place (read before section)
- **Lecture 2 (this week):** GitHub, LinkedIn, and public artifacts
- **Section (this week):** Resume teardown clinic, three live bullet rewrites, sketch one public artifact
- **HW 1 (out week 1, due week 3):** Resume and portfolio audit; the section work feeds directly into this